

Chapter 1

Introduction

This work is concerned with tipping points, also referred to as critical transitions, which are characterised by Kuehn [2011] as follows:

A non-mathematical working definition of a critical transition is an abrupt change in a dynamical system.

Scheffer [2009] uses the example of a canoe capsizing as the canoeist leans further over the ~~non-~~ edge. This example is one in which the *dynamical system* (the status of the roll of the canoe as it changes over time) has two stable states (upright and upside-down) and transitions *abruptly* from one to the other as a critical threshold is reached, in this case the point at which the centre of mass is sufficiently offset from the canoe's axis of vertical symmetry that it becomes unstable. *not necessarily*
abrupt transitions are often
bifurcational

✓✓ In order to take this thesis beyond a superficial study of the canoeist example we introduce, in sections 1.1 and 1.2 ~~of this introductory chapter~~, the concepts of dynamical systems, time series analysis and tipping points, including some common examples. In section 1.3 we introduce the foundational ideas of early warning signals, which may be used to predict tipping points. In section 1.4 we outline the structure and key points of the remainder of this thesis.

1.1 Dynamical systems and time series analysis *other disciplines*

✓✓ The term *dynamical system* is used in ~~pure~~ *mathematical* analysis, medicine [Jackson and Radunskaya, 2015], *climatology* and ~~everywhere in between~~. For the purposes of the physical sciences a dynamical system is defined as "a particle or ensemble of particles whose state varies over time" [Nature, 2019]: *? Suriname* that is, almost any measurable quantity or quality of almost anything. The mathematical field of Dynamical Systems ~~might~~ more specifically define a dynamical system as a triple (S, T, Φ) , *changing over time*

where S is the state space, $\mathcal{T}$ is the parameter space and the map

$$\Phi : (S \times \mathcal{T}) \rightarrow S \quad (1.1)$$

is the evolution of the system, ~~the function~~ determining the dependence of the state $s \in S$ on the parameter $t \in \mathcal{T}$ [Brown, 2018]. The state of the system as the parameter changes is called the *trajectory* of the system and $(S \times \mathcal{T})$ is known as the trajectory space.

The majority of the influence on this thesis is tipping point research relating to real physical systems changing over time, such as the change in atmospheric oxygen concentration over a period of years [Livina et al., 2015]. In chapter 4 the tipping point analysis techniques developed and studied throughout the previous chapters are applied to observed sea-level pressure measurements as they change over a period of days. Therefore, in the context of this thesis, $t \in \mathcal{T}$ is typically the "time" parameter and $\mathcal{T}$ is typically the set of real numbers $\mathbb{R}$, or a discrete subset (e.g. $\{0, 0.1, 0.2, \dots\}$). Hence, the trajectory is the state of the system as it changes with time. This brings the mathematical definition of a dynamical system closer to the physical sciences definition. In this thesis we generally distinguish an analytically defined *dynamical system*, where the map Φ is often defined by a set of differential equations [Robinson, 2012], from a *physical system*, such as the roll angle of a canoe or the sea-level pressure at a specific geographical location. However, the two concepts overlap where a physical system is presumed to be governed by (possibly unknown) analytical equations, or when an analytical dynamical system is used to model the behaviour of a physical variable. A simple example of a dynamical system is based on the quadratic map

$$f(x) = 4x(1 - x). \quad (1.2)$$

Here, the state is $x \in \mathbb{R}$, the 'time' parameter is $n \in \mathbb{N}$ and the dynamics are governed by the map

$$\Phi : (x, n) \rightarrow f^n(x), \quad \text{lower index?} \quad (1.3)$$

denoting the function f iterated n times. The trajectory of this dynamical system, with the initial condition $x_0 = 0.4$, is shown in figure 1.1. The series of values $X = \{x_0, x_1, \dots\}$ is, if we continue to refer to the parameter n as 'time', a *time series*, a collection of observations made over time [Chatfield, 2016]. These observation can be the price of a commodity, the yearly global average temperature deviation or, in this case, the state x of the dynamical system in equation 1.3. *hourly*

In chapters 2 and 3 several dynamical systems defined by analytical equations are studied in order to detect or predict tipping points which occur in those systems. An instructive example,

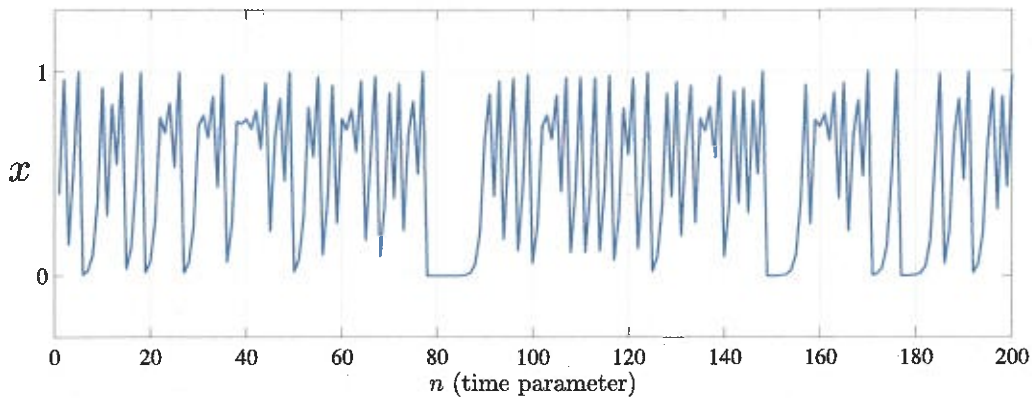


Fig. 1.1 The quadratic map (equation 1.2) iterated 200 times with initial value $x_0 = 0.4$.

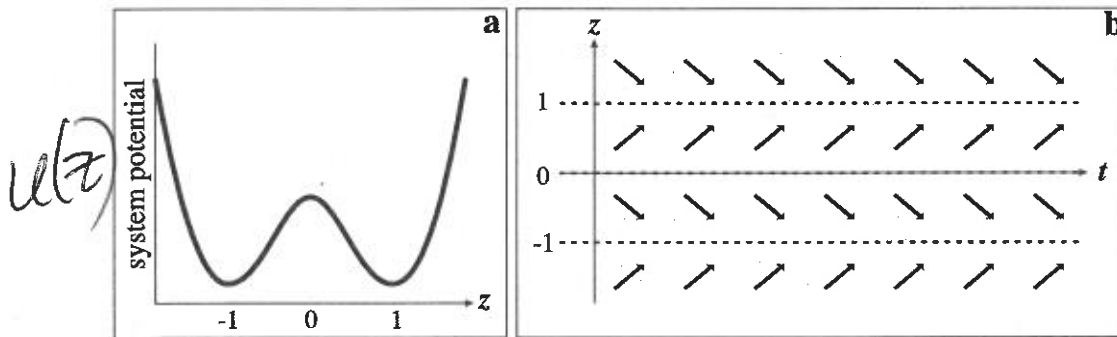


Fig. 1.2 (a) The shape of the double-well potential function $U(z) = z^4 - 2z^2$, and (b) the trajectory of the dynamical system described by equation 1.4. The dashed lines represent the equilibria $z = -1, 0, 1$.

which is referred to throughout this thesis, is the *double-well potential* system, described by the differential equation

$$\dot{z} := \frac{dz}{dt} = -\frac{\partial}{\partial z} (z^4 - 2z^2), \quad (1.4)$$

where z is the system state variable and t is the time parameter. The RHS could be written more concisely as $4z - 4z^3$ but expressing it as a derivative allows us to see the shape of the *potential function* $U(z) = z^4 - 2z^2$, which is illustrated in figure 1.2a. As t changes, so to does the system state z unless $\dot{z} = 0$, which occurs at the *equilibria*, the states $z = -1, 0, 1$, which are also the solutions to

$$\frac{\partial}{\partial z} (z^4 - 2z^2) = 0. \quad (1.5)$$

Thus, we are able to understand the dynamical system in terms of its equilibria. At $z = 0$ there is an *unstable equilibrium* where $U'' = 12z^2 - 4 = -4 < 0$ and so if z is close to, but not equal

to zero, the evolution of the system (the change as t increases) will move away from this state, along the arrows of the trajectory in figure 1.2b, towards the *stable* equilibria $z = \pm 1$, where $U'' = 12z^2 - 4 = 8 > 0$. Because of the shape of the potential function, these stable equilibria are known as wells or basins of attraction. Any state $z \neq 0$ will evolve towards either one of these wells and will become asymptotically close to $z = \pm 1$.

When attempting to model a physical system, a purely deterministic equation may not be appropriate [Shumway and Stoffer, 2017] and so it is useful to couple a dynamical system with a stochastic process to model uncertainty or unknown short-scale processes. The stochasticity can also induce tipping in the dynamical system — we note that in the double-well potential system introduced above, the system variable z heads towards one of the potential wells (one of the stable states of the system) and remains there: this is not a *useful* model of a physical system such as the canoe example, which tips from one state to another. *suitable for*

The most basic example of a stochastic process, for our purposes, is independent Gaussian white noise, that is, a time series where each element is sampled independently from a zero-mean Gaussian distribution. In several systems throughout this thesis we introduce additive noise [Honerkamp, 1994] so that a system

$$\frac{dx}{dt} = f(x), \quad (1.6)$$

becomes

$$\frac{dx}{dt} = f(x) + \eta_t, \quad (1.7)$$

where η is a Gaussian white noise process. Adding a white noise process to the double well potential system in equation 1.4, we get a stochastic dynamical system described by the stochastic differential equation

$$\dot{z} = -\frac{\partial}{\partial z}(z^4 - 2z^2) + \sigma\eta_t, \quad (1.8)$$

where the constant σ alters the variance of the noise term. In this system the overall evolution is still towards two wells of attraction, we note that a particle starting exactly in the state $z = 0$ will no longer remain on the unstable equilibrium as perturbations from the noise term will move the particle slightly away from zero, after which point it will be attracted into one of the two wells. In figure 1.3 the trajectory of the system is shown with initial condition $z = 0$ for small noise ($\sigma = 0.4$) and for large noise ($\sigma = 0.7$). Note that when the noise is sufficiently large a particle is able to cross over the $z = 0$ point, the boarder between the two wells of attraction, and move into the other well. This noise-induced transition from one stable state to another is one of the several types of tipping, which will be described in section 1.2. *Note that*

this transition is abrupt and non-bifurcational (the shape of the system potential does not change).

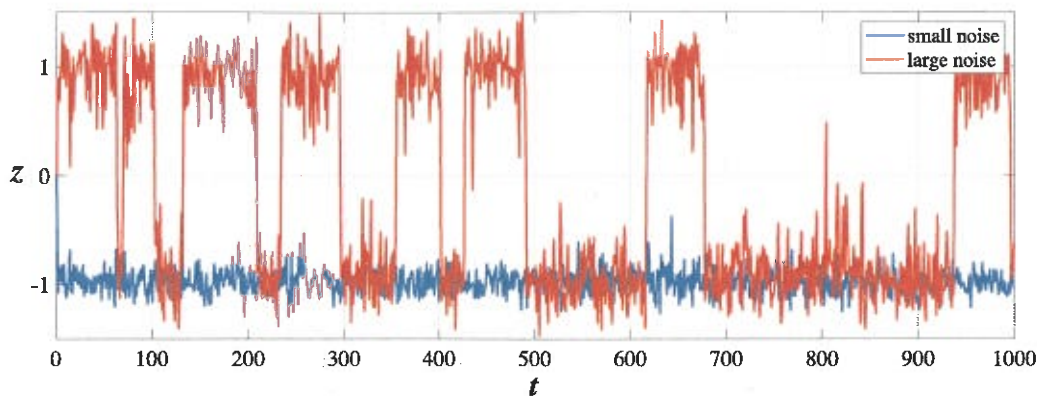


Fig. 1.3 Time series of the possible trajectory of the stochastic double well potential system described by equation 1.8 with small noise (blue line, $\sigma = 0.4$) and large noise (red line, $\sigma = 0.7$).

Besides Gaussian white noise, we may introduce many different types of stochastic processes: Brownian motion, fractional ARIMA processes, the Ornstein-Uhlenbeck process, etc. [Box et al., 2008; Chatfield, 2016; Shumway and Stoffer, 2017]. We make a note of the *discrete random walk*, which is simply the cumulative sum of a white noise series, thus:

$$x_t = \sum_{j=1}^t \eta_j, \quad (1.9)$$

where η is Gaussian white noise. This could also be defined recursively by the equation

$$x_t = x_{t-1} + \eta_t, \quad (1.10)$$

with some initial condition $x_0 = 0$. A more general stochastic process with a similar form is the order- p autoregressive process, or AR(p) process, is described by the equation

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + \eta_t. \quad (1.11)$$

where $\phi_1, \phi_2, \dots$ are constant parameters. Figure 1.4 shows the time series of a random walk, and an AR(2) process with $\phi_1 = 0.8, \phi_2 = -0.2$. These two time series appear different, we note that the AR(2) process appears to be stationary (the mean does not change over time) whereas the random walk does not.

There exist many methods for the analysis of the time series which arise both from measurements of physical systems and from analytically defined dynamical systems [Box et al., 2008; Chatfield, 2016]. Besides the mean and sample variance, we define the sample autocorrelation

high value, you get some nonstationarity
on fig 1.4.
try smaller value for
distinct difference with random walk

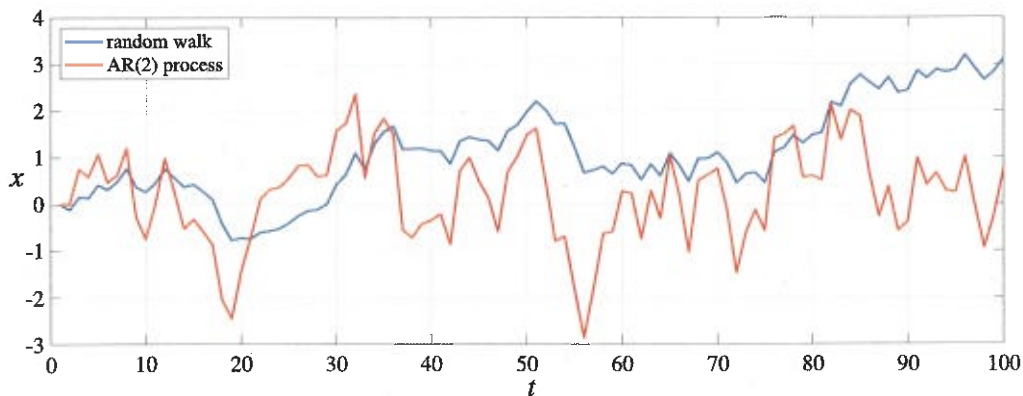


Fig. 1.4 Time series of a random walk (blue line), and an AR(2) process with $\phi_1 = 0.8$, $\phi_2 = -0.2$ (red line).

function (ACF), a measure of the correlation between a time series and a delayed copy of itself. Since, in tipping point research, we are concerned with abrupt changes in systems, autocorrelation is frequently used to analyse time series of systems in which tipping has occurred or may occur [Held and Kleinen, 2004; Scheffer et al., 2009]. In chapter 2 the ACF indicator is introduced as a means to quantify how autocorrelation changes over time.

The ACF with lag l of a time series X is given by

$$\text{ACF}_l(X) = \frac{1}{(N-l)s^2} \sum_{j=1}^{N-l} (X_j - \bar{X})(X_{j+l} - \bar{X}), \quad (1.12)$$

numerically

where $\bar{X}$ and s are the mean and standard deviation of X . The lag-1 ACF of the random walk and the AR(2) process in figure 1.4 are found *empirically* to be 0.9995 and 0.6658 respectively. The ACF of a Gaussian white noise process is zero, since each point is sampled independently and so there is no correlation between them.

Many methods of time series analysis involve transforming the series from the *time domain*, in which the system state is a function of time, into the *frequency domain*, in which the amount of the series lying within a frequency band is a function of frequency [Chatfield, 2016]. By *applying* using the fast Fourier transform [Cochran et al., 1967] *to* a discrete time series to represent the series as a sum of sine waves, we are able to plot the periodogram, showing the spectral power, or relative amplitude of the sine wave, at each frequency. The periodogram is a discrete approximation to the power spectrum.

Figure ?? shows the periodogram of a Gaussian white noise series and a random walk (both of length 10^6). Because each point of the white noise series is independent, all frequencies occur with equal likelihood and the power spectrum is flat. The random walk has greater power

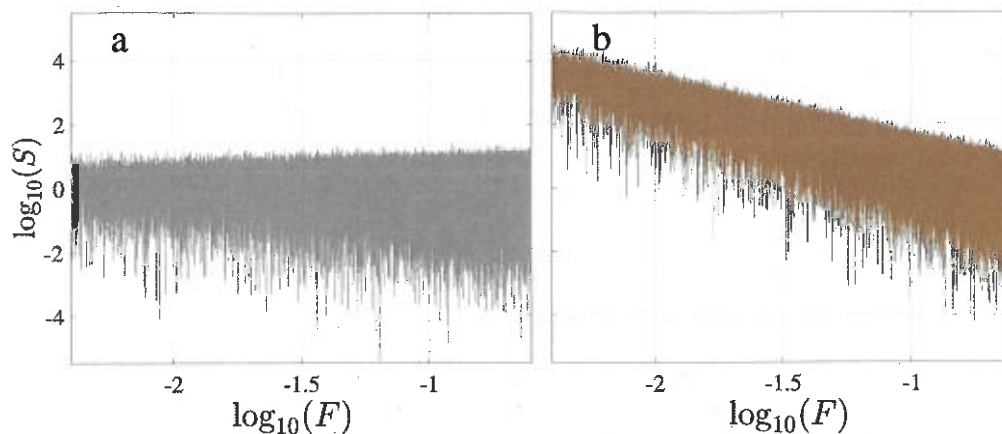


Fig. 1.5 The power spectrum of a Gaussian white noise series (panel a) and a random walk (panel b) both of length 10^6 . The logarithm of the spectral power S is plotted over the logarithm of the frequency F in both cases in the frequency range $10^{-2} < F < 10^{-1}$.

in the lower frequencies because points close to each other are more strongly correlated. Thus, measuring the slope of the power spectrum gives, in some respect, similar information to the calculation of the autocorrelation function, and we see that the power spectrum and the ACF are closely related (see the Wiener–Khinchin theorem, Champeney [1989]). This allows us, in chapter 2 to introduce a method for detecting tipping points based on analysis of the power spectrum, using the knowledge that the ACF is already widely used for this purpose. Also in chapter 2, we describe yet another time series analysis technique: detrended fluctuation analysis [Kantelhardt et al. [2001]], and we provide a comparison of the different techniques for early warning signals, which are based on scaling properties of time series.

The time series and dynamical systems described above have only a single-valued system variable x , which is required by techniques such as the ACF or the power spectrum. In chapter 3 we expand the scope of the thesis to include systems such as

$$\begin{aligned} \dot{r} &= \mu r - r^3, \\ \dot{\theta} &= 1 + r^2, \end{aligned} \quad (1.13)$$

which has two variables (in this case, r and θ). In a physical system, this could correspond to measuring several different physical variables, or measuring the same variable at several discrete locations in space. It may be possible only to consider a single critical variable such as temperature or oxygen concentration [Dakos et al., 2008]. Otherwise it is also possible to combine several time series into one, for example using EOFs¹, which we discuss in section 3.2,

¹Empirical Orthogonal Functions

✓ or to use a technique similar to ACF but expanded to higher dimensions [Williamson and Lenton, 2015], which we discuss in section 3.1. As a further alternative we introduce, in section 3.6, the possibility of visualising several time series simultaneously. All of these ideas are then applied to a practical problem in chapter 4.

1.2 Bifurcations and tipping points in dynamical systems

✓ In this section we introduce the concept of a bifurcation, which will be closely related to the concept of a tipping point. Introduced by Poincaré and meaning a split or fork, the general term bifurcation does not have a specific meaning, although specific types of bifurcation are well defined. It is understood to mean a change in the parameters of a dynamical system which causes a significant qualitative change in the system's behaviour [Devaney, 2003; Blanchard et al., 2006]. The dynamical systems and stochastic processes seen in the previous chapter do not contain bifurcations in any established sense although the change in the parameter (time) does cause changes in the systems. In the double-well potential system in figure 1.3 (large noise) switches from one well to another but this cannot be called a bifurcation since the change occurs due to perturbations from the noise term, not due to a change in the parameter t (that is, it could potentially happen at any time t). We can introduce a bifurcation into the system by varying the coefficient of z^2 in equation 1.8 so that the shape of the potential function U changes:

$$U = z^4 + \left(3 - \frac{t}{200}\right) z^2. \quad (1.14)$$

When $t = 1000$ we have the same double-well potential system as in equation 1.8 with $U = z^4 - 2z^2$. However, when $t < 600$ the shape of the potential function is qualitatively different: only a single potential well exists. Figure 1.6 shows the evolution of U over time and the time series in figure 1.7 show two possible trajectories of the system. The bifurcation at $t = 600$ is another example of a tipping point, distinct from the noise-induced transition seen in figure 1.3. In chapter 2 this same dynamical system is used to create multiple time series on which we test the tipping point techniques described in that chapter and successfully identify the tipping event at $t = 600$.

The casual definition of a tipping point offered by Kuehn [2011], "an abrupt change in a dynamical system", suggests that tipping points are not restricted to a particular type of change in a dynamical system, either a bifurcation or a transition from one state to another, but any system in which a sudden change can be observed.

Because the definition is very broad, tipping points are studied in a variety of different contexts using a variety of different methods [Scheffer, 2009]. Some areas of research in which tipping points are studied include finance [Scheffer et al., 2009], ecology [Scheffer

analysing

using 2D plots

geophysical

of branching

in
primer
dynamical
systems

Bifurcation in Fig 1.7 is not actually abrupt

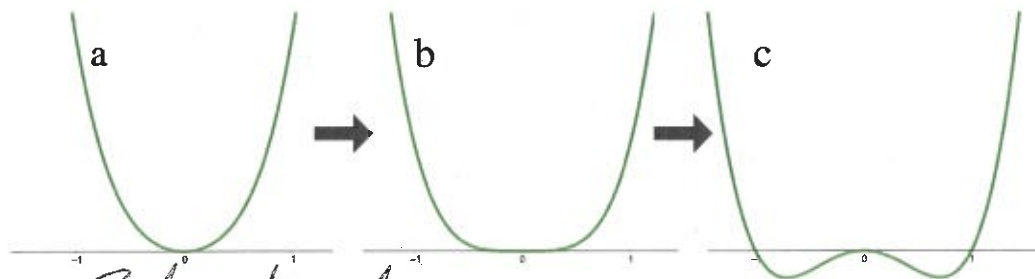


Fig. 1.6 Illustrating the changing shape of the potential function U in equation 1.14. U is shown for $t < 600$ where the shape is a "single well", for $t = 600$ where the bifurcation occurs; and for $t > 600$ when the "double well" shape has developed.

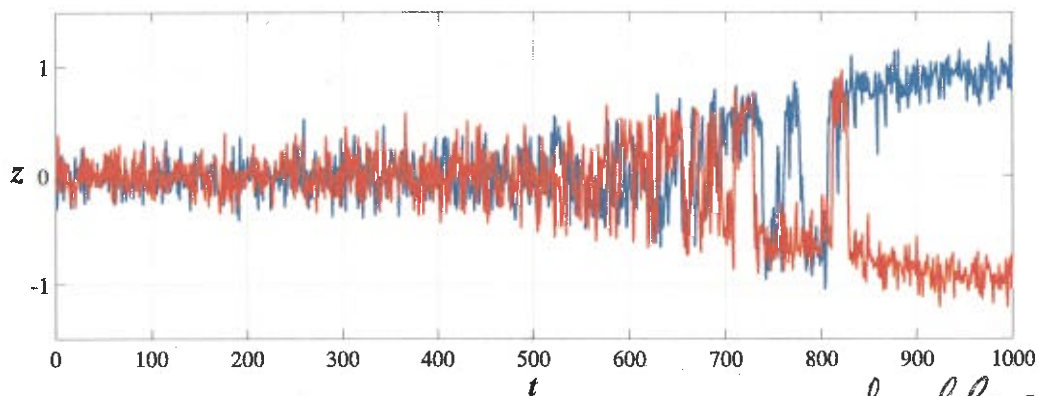


Fig. 1.7 Time series of two possible trajectories of the stochastic double-well-potential system with changing potential function described by equation 1.14. The equation is the same in both cases, with noise level $\sigma = 0.4$.

mathematically

et al., 2001], biochemistry [Bagowski and Ferrell Jr, 2001] and climate [Lenton et al., 2008]. Despite the breadth of the subject, many systems exhibiting tipping points are related, that is, when these systems are close to the tipping point they share certain general properties in their dynamics which arise regardless of their individual natures [Scheffer et al., 2009]. These shared properties, enumerated by Scheffer [2009] are summarised by Kuehn [2011] in the following list:

1. An abrupt qualitative change in the dynamical system occurs.
2. The change occurs rapidly in comparison to the regular dynamics.
3. The system crosses a special threshold near a transition.
4. The new state of the system is far away from its previous state.
5. There is small noise in the system i.e. the data has a major deterministic component with small "random fluctuations".
6. The system recovers slowly from perturbations ("slowing down").
7. The variance of the system increases as the transition is approached.
8. The noisy fluctuations become more asymmetric.
9. The autocorrelation increases before a transition.

Vague?

The point of interest is the third point: "The system crosses a special threshold near a transition." since the nature of the "special threshold" determines the nature of the tipping point and therefore which techniques are used to analyse it.

The idea of the critical threshold comes from catastrophe theory which has in the past been criticised for an unscientific application of the cusp catastrophe to a huge variety of social and biological systems [Zahler and Sussmann, 1977], including the fight reflex of dogs and the psychological states of *anorexia nervosa* sufferers [Zeeman, 1977], all of which represent tipping points. Despite this, the cusp catastrophe is widely used as an example in tipping point research [Scheffer et al., 2001; van Nes and Scheffer, 2005; Van Nes and Scheffer, 2007; Kuehn, 2011]. Scheffer et al. [2009] note that the cusp model is "now considered to capture the essence of shifts at tipping points in a wide range of natural systems" and also that critical thresholds for tipping points correspond to bifurcations. In figure 1.8 three tipping points are illustrated: a non-catastrophic transition in which the system state changes smoothly; a cusp catastrophe in which the system undergoes a bifurcation; and a transition from one stable state to the another. In the first two cases (panels a and b) the same forcing is used and the same final system state

in broad sense.

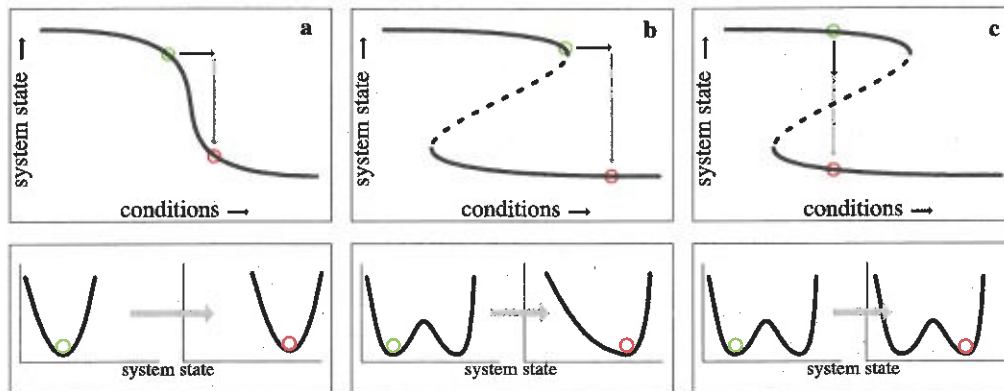


Fig. 1.8 Three distinct tipping points: a non-catastrophic transition (a), a cusp catastrophe bifurcation (b) and a transition between stable states (c). In each case the small black arrow represents a small forcing and the grey arrow represents the resulting change in the system state. The attractor is shown in the top panel and the system potential function is shown beneath.

is reached. Given the a sufficiently large forcing, therefore, it would not be possible to tell the difference between the two scenarios. We use this idea in chapter 4 to justify the use of ~~traditional~~ tipping point analysis in a novel system: the approach of a tropical cyclone. This system is possibly not an example of a two-state system with a genuine bifurcation or transition, but it is possible that the same analysis will work since, ~~possible~~, the same sort of forcing is present as if it were a bifurcation. Indeed, Ashwin et al. [2012] note, in the context of tipping point analysis, that tipping points do not only include bifurcations.

A comparison of the transitions in figure 1.8 panels b and c demonstrates the two possible ways to shift between alternate stable states [Scheffer et al., 2001]: either a slight change in the system conditions, possibly a parameter value, close to the bifurcation point; or a sufficiently large perturbation due to noise which is able the push the system over the boarder to the other attractor well. Ashwin et al. [2012] describe three types of tipping with applications to climate: bifurcations, noise-induced tipping, and rate-induced tipping. Livina et al. [2011] identify three types of system changes which may be the cause of a tipping event:

1. A forced transition, where the potential function stays the same shape, but is moved, so that there is a trend (or possibly a sudden jump) in the time series.
2. A genuine bifurcation, where the shape of the potential function is changed.
3. A noise-induced transition, where the system shifts from one stable state to another due to a large initial perturbation.

unless, in case (b), the second state is sampled by the system, combining (b) and (c)

We note that these three scenarios are represented in figure 1.8 (a, b and c resp.) where the forcing of the system state in figure 1.8c is ~~considered~~ ^{- driven} noise. Rate-induced tipping, which is not represented in figure 1.8, occurs as a result of the rate of change of a system parameter (or system conditions).

Our working definition of a tipping point refers to an abrupt change in a dynamical system, which may also refer to a change in the system potential or underlying mechanics, besides simply the measurable state of the system as the three examples in figure 1.8 suggest. Livina et al. [2011] study a system such as

$$\dot{z}(t) = -\frac{\partial}{\partial z}(z^4 - 2z^2) + \sigma(t)\eta, \quad (1.15)$$

where η is white noise and $\sigma(t)$ is a parameter which varies with time and governs the size of the noise. In this example, where $\sigma(t)$ is made to decrease from 2 to close to zero, a system with a double-well potential effectively becomes a single-well system as the noise term becomes so small that it is practically impossible for the system to change state, although the actual shape of the potential function does not change. If the change is sufficiently sudden compared to the long-term dynamics, this system could be said to exhibit a tipping point at around the time that $\sigma(t)$ starts decreasing, even though the system state (the value of z) may not apparently change at that point.

1.3 Early warning signals of tipping points

Closely related to the idea of tipping points is the idea of early warning signals, since it is often useful to be forewarned of an impending catastrophe. The above list of nine shared tipping-point attributes references an "increase in variance" and an "increase in autocorrelation" prior to the tipping point, both of which facts are commonly used to predict a tipping point before it happens, given that the existence of a critical threshold is known [Scheffer et al., 2009; Thompson and Sieber, 2011]. Thus, it is possible to use measurements of variance and autocorrelation as tipping point indicators which may provide early warning signals of a future event. This is basis of all the techniques used throughout this thesis, although we also use measurements of the power spectrum properties which is closely related to autocorrelation, and detrended fluctuation analysis (DFA), both of which are described in section 2.1. In order to see the change in these statistics over time we use a tipping point indicator [Held and Kleinen, 2004]: the ACF, variance, ~~etc~~ ^{and others} measured in overlapping segments of the time series. In chapter 2 three tipping point indicators (ACF, DFA, and the novel PS indicator) are applied to various time series.

This idea assumes that the change is rapid but not abrupt

and others are

Held & Kleinen
Livina & Lenton

this is equivalent
to increasing memory correlations
or long-term correlations

A key concept involved in these methods is that of resilience or recovery rates [Scheffer et al., 2001; Veraart et al., 2012], that is, how quickly the system returns to the stable state having received a small perturbation. The theory states that the recovery rate will slow down as the system approaches a bifurcation point [Scheffer et al., 2009], known as critical slowing down. As an example we consider the pitchfork bifurcation illustrated in figure 1.6. This system is trapped within a single well of attraction (at $z = 0$) for $t \leq 600$, at $t = 600$ a bifurcation occurs creating a double well potential system for $t > 600$. for $t \ll 600$ the sides of the potential well are steep and the system quickly returns to the equilibrium $z = 0$ following a small perturbation due to noise. For $t = 600$ (panel b), however, this process takes much longer because the base of the well is flat. During the transition from panel a to panel b the system will take longer and longer to return to the $z = 0$ equilibrium following a small perturbation, as the sides of the potential well become less steep. This increasing return time, or decreasing resilience is known as critical slowing down. Because of the generalness of the pitchfork bifurcation [Kuehn, 2011], the hypothesis is that critical slowing down will be observed in a wide variety of dynamical systems approaching a bifurcation [Scheffer et al., 2009]. This is tested by Van Nes and Scheffer [2007] who measure critical slowing down in a number of different ecological models and find the effect to occur universally.

The speed at which a system returns to equilibrium can be measured by the autocorrelation, since a non-tuning system will behave as a random walk, ~~and a system which always returns quickly is governed only by noise~~ [Scheffer et al., 2009]. It is this observation which allows us to successfully apply autocorrelation-based tipping point indicators to several different dynamical systems in chapters 2 and 3, and to use these indicators to detect tipping points in time series of geophysical variables in chapter 4.

1.4 Outline of thesis

In chapter 2 we study different tipping point indicators which are applied to one-dimensional time series exhibiting tipping points. The indicators include the lag-1 autocorrelation function [Held and Kleinen, 2004; Dakos et al., 2008], the use of which has been mentioned in this introduction, the detrended fluctuations analysis (DFA) coefficient [Kantelhardt et al., 2001], which has been used previously in tipping point research [Livina and Lenton, 2007], and the power spectrum scaling exponent, the use of which in tipping point research is a novel outcome of this thesis [Prettyman et al., 2018].

In chapter 3 the tipping point indicators used in chapter 2 are adapted for use in multi-dimensional systems, both by evaluating the indicators simultaneously at many points over a field to provide a visual description of the indicator's behaviour, and also by reducing the

dimension of a system using empirical orthogonal functions (EOFs) [Held and Kleinen, 2004]. The specific applicability of the EOF method to tipping point analysis is investigated analytically using a general dynamical system containing a tipping point, and experiments are also made with novel variations of the EOF method in a tipping points context. The methods presented are also compared to an existing method for producing early warning signals of tipping points in two-dimensional systems [Williamson and Lenton, 2015].

In chapter 4 the methods introduced throughout the thesis are applied to measurement data from a physical system, that is, to measurements of sea-level pressure at points close to the landfall locations of tropical cyclones. A stochastic model of an approaching tropical cyclone is also presented, adapted from a simple existing model [Holland, 1980] and parametrised using the results from our tipping point analysis.

geophysical

modified